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Clinical Approaches to Immunological Deficiencies

Adam Hurst, M.D.
Clinical Medicine
ahurst@nyit.edu

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Session Objectives

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1. Explain the difference between primary and secondary immunodeficiency (ID) and outline some causes of secondary ID.
2. Explain the signs and symptoms that should raise suspicion for a primary ID (PID).
3. Identify the history and physical exam components important when considering an ID.
4. For the following PIDs, explain the underlying immune defect, the infectious clinical features, the non-infectious clinical features, laboratory features and the main treatment approaches:
   - Common Variable Immunodeficiency
   - Selective IgA Deficiency
   - IL-12 Receptor Deficiency
   - Chronic Mucocutaneous Candidiasis
   - Chédiak-Higashi Disease
   - Chronic Granulomatous Disease

Covered in this lecture:

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Diagnosed in Infancy	Diagnosed after Infancy
• X-linked agammaglobulinemia • Sever combined immunodeficiency (SCIDS) • Ataxia – telangiectasia • Wiscott-Aldrich Syndrome • Leukocyte adhesion deficiency (Type 1) • Hyper IgM Syndrome (Job syndrome)	• Selective IgA deficiency • Common variable immunodeficiency • IL-12 receptor deficiency • Chronic Mucocutaneous Candidiasis • Chédiak-Higashi syndrome • Chronic Granulomatous Disease

Primary vs Secondary Immunodeficiency (ID)

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Definition: defect in the immune system resulting in a weakened OR dysregulated immune defense

Usually presents as unusual, recurrent or prolonged/severe infection

Primary ID:

- Inherent Defect in Immunity
- Most arise from inborn deviation in genetic code
- Usually presents in infancy or childhood, rarely adulthood
- Problems with B cells are most common PID (>50%)

Secondary ID:

- Medications – immunosuppressants, systemic steroids
- Viral infections – HIV, measles, rubella
- Protein losing states – nephrotic syndrome
- Metabolic disorders – diabetes, malnutrition, vitamin deficiency
- Other: prematurity,, acquired asplenia, acquired neutropenia from drugs or virus, autoimmune disorder (ex. SLE), malignancy

Features of PID – more than infections

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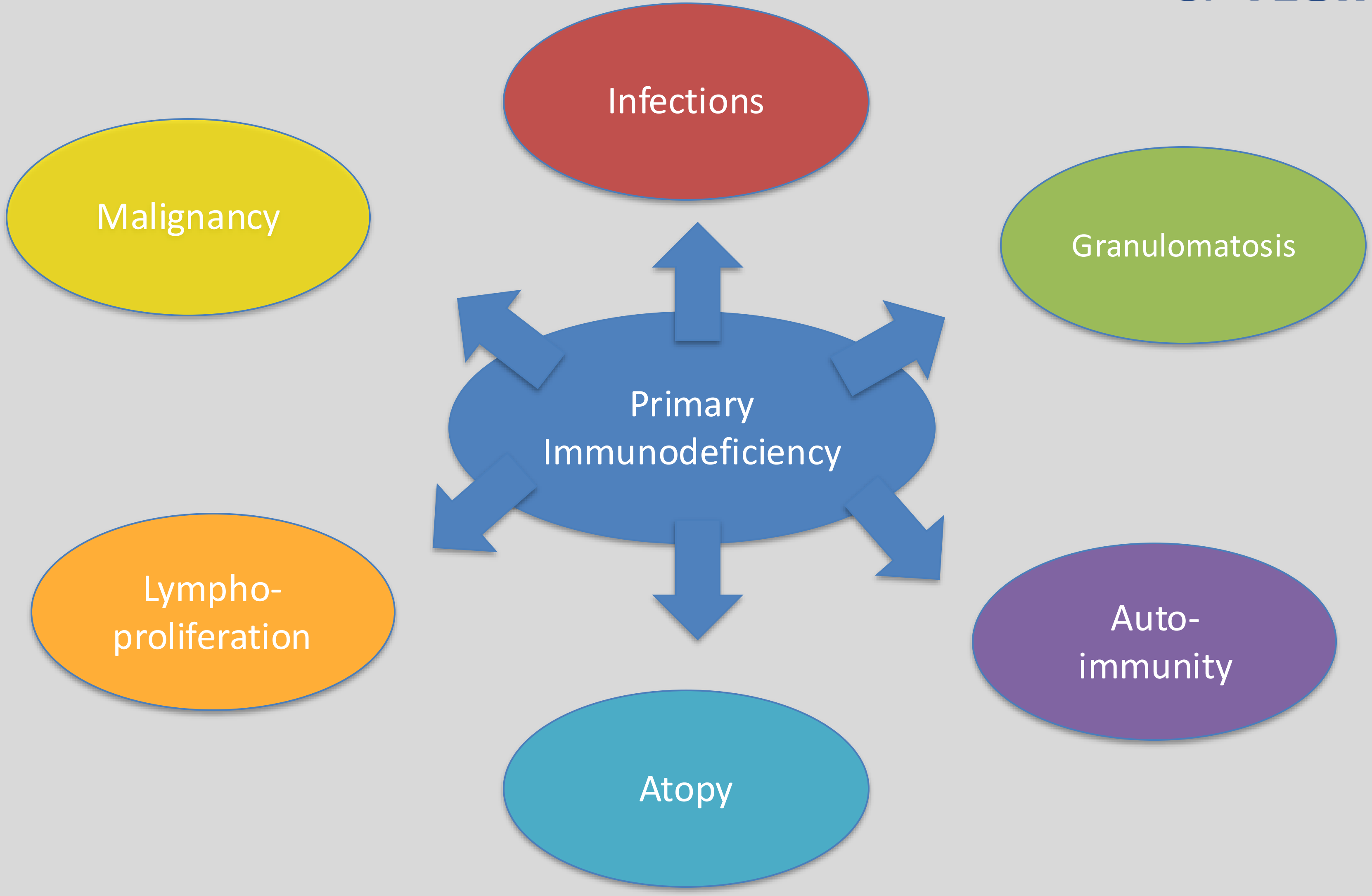
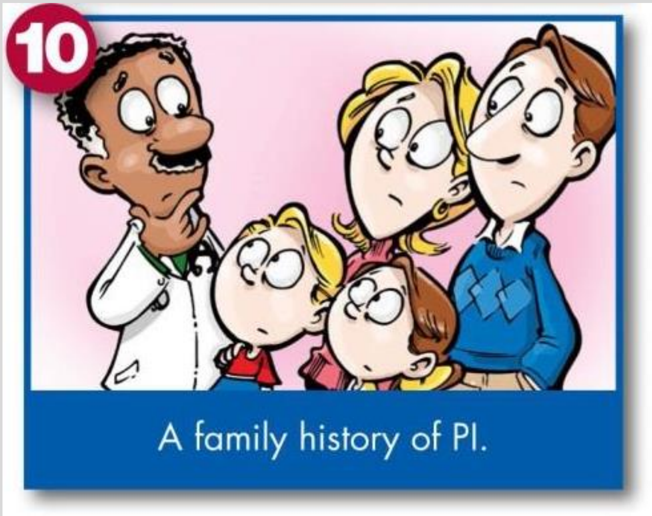


Adapted from : Immunol Allergy Clin N Am 35 (2015) 599-623

When to Suspect a PID

https://www.info4pi.org/library/educational-materials/10-warning-signs

When to Suspect a PID



Other clues:

- Serious infections at unusual sites
- Infections with unusual pathogens
- Unusually severe infections with common pathogens

https://www.info4pi.org/library/educational-materials/10-warning-signs

Important History Items

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- Birth history
- Growth and development
- Immunizations – adverse effects, failures
- Medications –duration, effectiveness, adverse effects, immunosuppressants
- PMH – FTT, noninfectious disorders
- Family History – of PID, death of young child due to infection, consanguinity
- Social history – environmental exposures
- Infections: age, location, frequency, severity, organism, response to treatment
- Comprehensive ROS

Important Physical Exam Items

- Height and weight: evaluate growth chart and trends
- HEENT: dysmorphic features, nasal discharge, oropharynx - thrush, aphthous ulcers, gingivitis
- Lymph nodes
- Abdomen: hepatosplenomegaly
- Musculoskeletal- scoliosis, multiple fractures, arthritis
- Dermatologic- hair pigmentation and thickness, scars, telangiectasias, eczema, warts

*Physical exam may be normal

Work-up for suspected ID

- History and physical exam
- CBC with differential counts
- Cultures/Imaging
- IgG, IgA, IgM, IgE levels
- Diphtheria, tetanus titers
- Pneumococcus, *Haemophilus* titers
- CH50, AH50

Step-wise approach

- Start with basics
- Additional tests depend on the suspected underlying defects

Sullivan, K, and Buckley, R., Evaluation of Suspected Immunodeficiency, Nelson Textbook of Pediatrics. (148):1097-1103. Figure 148.2.

Categorization of PID

B Cell	T Cell	Combined B and T Cell (CID)	Phagocytic
Common Variable Immunodeficiency	Job Syndrome (AD Hyper-IgE Syndrome)	Ataxia Telangiectasia	Leukocyte Adhesion Deficiency Type 1
Selective IgA Deficiency	Thymic Aplasia	Hyper IgM Syndrome	**Chediak-Higashi Syndrome**
X linked Agammaglobulin-emia	**IL-12 Receptor Deficiency**	Wiskott-Aldrich Syndrome	**Chronic granulomatous disease**
	Chronic Mucocutaneous Candidiasis	Severe Combined Immunodeficiency	

Typical Infections by Category of PID

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TABLE I. Type of infections associated with major categories of PIDs

Organism	Antibody deficiencies	CIDs	Phagocytic defects	Complement deficiencies
Viruses	Enteroviruses	All, especially: CMV, respiratory syncytial virus, EBV, parainfluenza type 3	No	No
Bacteria	*Streptococcus pneumoniae, Haemophilus influenzae, Moraxella catarrhalis, Pseudomonas aeruginosa, Staphylococcus aureus, Neisseria meningitidis, Mycoplasma pneumoniae*	As for antibody deficiencies, also: *Salmonella typhi*, *Listeria monocytogenes*, enteric flora	*S aureus, P aeruginosa, Nocardia asteroides, S typhi*	As for antibody deficiencies: especially *N meningitidis* in deficiency of late components
Mycobacteria	No	Nontuberculous, including BCG	Nontuberculous, including BCG	No
Fungi	No	*Candida* species, *Aspergillus* species, *Cryptococcus neoformans, Histoplasmosis capsulatum*	*Candida* species, *Aspergillus* species	No
Protozoa	*Giardia lamblia*	*Pneumocystis jiroveci, Toxoplasma gondii, Cryptosporidium parvum*	No	No

CID – combined immunodeficiency (B & T cell) J Allergy Clin Immunolo 2010; 125:S182-94

Work-up for suspected ID

- History and physical exam
- CBC with differential counts
- Cultures/Imaging
- IgG, IgA, IgM, IgE levels
- Diphtheria, tetanus titers
- Pneumococcus, *Haemophilus* titers
- CH50, AH50

If normal, consult with clinical immunologist, consider the following

Suspect phagocyte defect	Suspect humoral defect	Suspect combined defect
• Dihydrorhodamine for CGD	• Flow cytometry for T cells (CD4, CD8), B cells, NK cells	• Flow cytometry for T cells (CD4, CD8), B cells, NK cells
• Flow cytometry for LAD1 (CD11/CD18) or LAD2 (CD15)	• Enumerate percentage of naïve, memory, and switched memory B cells	• Enumerate percentage of memory vs naïve T cells (CD45 isoform expression)
• Consider neutrophil chemotaxis assays	• Functional B cell studies	• T cell mitogen studies
• IL-12 receptor/IFNγ receptor expression and function		

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Step-wise approach

- Start with basics
- Additional tests depend on the suspected underlying defects

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Work-up for suspected ID

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Table 148.7 Laboratory Tests in Immunodeficiency

SCREENING TESTS	ADVANCED TESTS	RESEARCH/SPECIAL TESTS
B-CELL DEFICIENCY		
IgG, IgM, IgA, and IgE levels	B-cell enumeration (CD19 or CD20)	Advanced B-cell phenotyping
Isohemagglutinin titers		Biopsies (e.g., lymph nodes)
Ab response to vaccine antigens (e.g., tetanus, diphtheria, pneumococci, *Haemophilus influenzae*)	Ab responses to boosters or to new vaccines	Ab responses to special antigens (e.g., bacteriophage φX174), mutation analysis
T-CELL DEFICIENCY		
Lymphocyte count	T-cell subset enumeration (CD3, CD4, CD8)	Advanced flow cytometry
Chest x-ray examination for thymic size*	Proliferative responses to mitogens, antigens, allogeneic cells	Enzyme assays (e.g., ADA, PNP)
TRECs	22q11.2 deletion analysis	Mutation analysis
		T-cell activation studies
PHAGOCYTIC DEFICIENCY		
WBC count, morphology	Adhesion molecule assays (e.g., CD11b/CD18, selectin ligand)	Mutation analysis
		Macrophage functional testing
Respiratory burst assay	Mutation analysis	
COMPLEMENT DEFICIENCY		
CH_{50} activity	AH_{50}, activity	Specific component assays

*In infants only.

Ab, Antibody; ADA, adenosine deaminase; C, complement; CH, hemolytic complement; G6PD, glucose-6-phosphate dehydrogenase; HLA, human leukocyte antigen; Ig, immunoglobulin; MPO, myeloperoxidase; NADPH, nicotinamide adenine dinucleotide phosphate; PNP, purine nucleoside phosphorylase; TRECs, T-cell receptor rearrangement excision circle; WBC, white blood cell; φX, phage antigen.

Sullivan, K, and Buckley, R., Evaluation of Suspected Immunodeficiency, Nelson Textbook of Pediatrics. (148):1097-1103. Figure 148.2.

B Cell Disorders

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- Usually no signs or symptoms until 3-4 months of age due to maternal IgG protection
- Antibodies are essential in sinopulmonary and mucosal surface protection
- Commonly present as recurrent otitis media, sinusitis or pneumonia but can have osteomyelitis, meningitis or sepsis
- Classic pathogens:
  - *Streptococcus pneumonia, Haemophilus influenza, Staph aureus, Pseudomonas aeruginosa, Mycoplasma* species
  - Chronic diarrhea from *Giardia lamblia, Salmonella enterica, Campylobacter jejuni*

B Cell
Common Variable Immunodeficiency
Selective IgA Deficiency
X linked Agammaglobulinemia

Common Variable Immunodeficiency (CVID)

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<u>Defect</u>: Heterogeneous group of disorders of Impaired Antibody Production

- Phenotypically normal B-cells do not differentiate due to an arrest in plasma cells differentiation
    - Results in hypogammaglobulinemia: Low IgG, Low IgA +/- Low IgM,
    - B cell numbers can be normal or low
- Various genetic defects and no specific genetic defect identified in most patients
- Defects can be due to AD, AR, or sporadic heritability
- Equally in males and females
- Usually presents in 2^{nd} – 3^{rd} Decade

Common Variable Immunodeficiency (CVID)

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Clinical Features:

Infectious

- Recurrent sinopulmonary infections: *Streptococcus pneumoniae, Haemophilus influenza, Mycoplasma*
- Chronic GI infections or malabsorption: *Giardia, Campylobacter, Salmonella, Helicobacter, Enterovirus*
- Can also have septic arthritis, meningitis or sepsis

Non-infectious

- **Features of Immune Dysregulation**
  - Autoimmune disease: RA, thyroiditis, autoimmune hemolytic anemia, alopecia, vitiligo, pernicious anemia
  - Malignancy: Increased risk of lymphoma (usually EBV-associated)
  - Granulomatous Disease: GI, liver, lungs
- Chronic Lung Disease: bronchiectasis, interstitial lung disease
- GI Disorders: IBD, protein-losing enteropathy

Common Variable Immunodeficiency (CVID)

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Labs:

- Markedly low serum IgG, low IgA and/or IgM
- Poor immunization response

Treatment:

- Antibiotics
- Immunoglobulin replacement therapy (IVIg)
- Monitor closely for noninfectious complications

Prognosis:

- Can have normal life expectancy if only infectious symptoms, but poor prognosis if autoimmunity or lymphoproliferation are present
- Autoimmune disease or lymphoproliferation confers poorer prognosis

Selective IgA Deficiency

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Defect: Isolated absence (or near absence) of serum/secretory IgA (<5-10 mg/dL).

- Underlying defect is unknown
- Phenotypically normal B cells

Clinical Features: Variable but Majority are asymptomatic,
The Most common Primary ID found in approximately 1 in 500 people

- **Infectious:**
  - Recurrent sinopulmonary infections: *Strep. pneumoniae, H. influenzae*
  - GI infections: *Giardia lamblia*
- **Non-infectious:**
  - Anaphylactic reaction to blood products/IVIg
  - Food or Respiratory allergies
  - GI disorders (Celiac disease, inflammatory bowel disease)
  - Increased incidence of auto-immune disorders

Selective IgA Deficiency

Labs: Cannot confirm until >4 years of age when IgA levels should reach adult levels

- Very low or absence of IgA levels (<5-10mg/dL)
- Normal IgM and IgG levels

Treatment:

- If asymptomatic – education of disorder and risks
- Treat recurrent infections as indicated
- If severe, screen for anti-IgA antibodies
- Good prognosis and most are asymptomatic, can progress to CVID

Question 1

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A 5-year-old female presents to your clinic with watery diarrhea and stomach cramps that have been present for multiple days. She has a PMH of a sinus infection and multiple GI problems within the past year and is up to date on immunizations with no adverse reactions reported. Stool antigen testing is positive for Giardia, and she is admitted to the hospital for IV fluids and recovery.

Once recovered, she returns to your office for follow up testing. A CBC and Ig panel show IgA of <5 mg/dL and normal IgM and IgG. What is the most likely diagnosis?

A. Common Variable Immunodeficiency

B. X-linked Agammaglobulinemia

C. Selective IgA Deficiency

D. Chronic Granulomatous Disease

Question 1

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C. Selective IgA Deficiency

D. Chronic Granulomatous Disease

T Cell Disorders

- Result in both Humoral and Cellular Immune Defects
  - Both are because T-cells and Antigen presenting cells provide necessary signals for B-cell differentiation
- Susceptible to bacterial, fungal, viral, mycobacterial and protozoal infections
  - Similar bacteria as with antibody deficiency
  - *Pneumocystis jiroveci, Mycobacteria, Cryptosporidium parvum Candidia, Aspergillus*
  - Viral infections often more severe, persistent, and disseminated

T Cell
Job Syndrome (AD Hyper-IgE Syndrome)
Thymic Aplasia
IL-12 Receptor Deficiency
Chronic Mucocutaneous Candidiasis

IL-12 Receptor Deficiency

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What is IL-12: cytokine secreted from macrophages, neutrophils and dendritic cells in response to infection with mycobacteria

- Binds to IL-12 receptor on NK cells and T-cells to stimulate IFN- ɣ secretion
- IFN- ɣ activates phagocytes to secrete TNF- α and destroy phagocytosed microbes

Defect: AR defect in either chain of the IL-12R resulting in loss of signaling

Clinical Features:

Infectious – Recurrent Infections: *Salmonella*, *Mycobacterium tuberculosis*, Nontuberculosis mycobacterium, BCG disseminated infection after vaccination

Non-infectious – not common

Typically presents in early childhood but even sooner in countries that administer BCG vaccine

IL-12 Receptor Deficiency

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Labs:

- Molecular testing – gene panels, sequencing candidate genes
- Immunologic testing – functional testing but not very useful

Treatment:

- Antibiotics to specific infection
- IFN-ɣ

Chronic Mucocutaneous Candidiasis

Defect: Homogeneous group of disorders characterized by an impaired immune response to Candida

- Defective Th17 lymphocytes or defective synthesis/signaling of IL-17
- Various gene defects: *AIRE* (autoimmune regulator) deficiency is AR, *STAT1* is AD

Clinical Features:

- Characterized by noninvasive chronic/recurrent candida infections of skin and mucus membranes
- Rarely develop invasive fungal disease due to other intact elements of the immune response
- Usually symptoms present in 1st month of life but can be as late as 2nd decade

Infectious Features:

- Chronic or recurrent Candida infections of skin, nails, mucous membranes
- Oropharyngeal candidiasis, intertrigo, onychomycosis
- Mainly *Candida albicans,* and, less frequently, by *C. glabrata*, *C. tropicalis*

Oral Thrush. (2020, July 8). Retrieved March 09, 2021, from https://www.nhs.uk/conditions/oral-thrush-mouth-thrush/

Chronic Mucocutaneous Candidiasis

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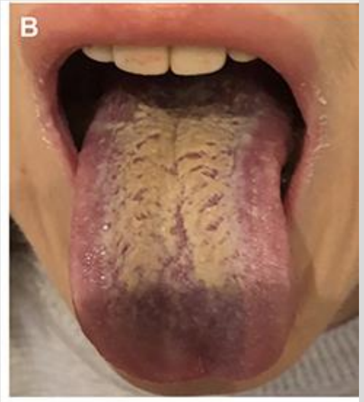
<u>Non-infectious Features</u>:

- Autoimmune polyendocrinopathy-candidiasis-ectodermal dystrophy aka APECED aka APS-1
    - *AIRE* defect
    - Recurrent Candida infections, autoimmune polyendocrinopathy (hypoparathyroidism, adrenal insufficiency and others), and skin dystrophy
- Other Autoimmune Disease: hemolytic anemia, ITP, neutropenia, RA
- Bone Marrow Failure: aplastic immunity
- Malignancy: mouth and esophagus



<u>Labs</u>: Primarily based on clinical features with genetic testing for underlying defect
Along with an endocrine work up if suspected

<u>Treatment</u>:

- Topical antifungals but often resistant
- Systemic azoles as required
- Address associated endocrine or autoimmune issues

https://www.frontiersin.org/files/Articles/723532/fped-09-723532-HTML/image_m/fped-09-723532-g003.jpg

Question 2

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A 4-year-old girl presents to the clinic with her mother for a recent nail infection that has not gotten any better with over-the-counter creams. She has a history of multiple infections of her skin and mouth, which usually respond to treatment with an antimicrobial cream that her mom cannot remember the name of. A picture of her nails is shown below. Which of the following is the most likely cause for this patient's symptoms?

A. Chediak-Higashi Syndrome
B. Chronic Mucocutaneous Candidiasis
C. Common Variable Immunodeficiency
D. IL-12 Receptor Deficiency

Question 2

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Phagocytic Dysfunction Disorders

- Phagocytic impairment can be due to inadequate neutrophil quantity **or** function
  - Neutrophils play an important role in the clearance of bacterial and fungal pathogens
- Primary ID due to defective phagocytic capacity is characterized by bacterial and fungal pathogens
  - Deep tissue infection, pneumonia, adenitis, sepsis, severe gingivitis, poor wound healing
  - *Staph aureus, Pseudomonas aeruginosa* can be life-threatening
- <20% of immunodeficiencies

Phagocytic
Leukocyte Adhesion Deficiency Type 1
Chediak-Higashi Syndrome
Chronic granulomatous disease

Chédiak-Higashi Syndrome

Defect: Disorder of intracellular granule formation

- AR mutation in CHS1/LYST
  - Involved with organellar protein trafficking
  - Abnormal fusion of vesicles and failure to transport lysosomes
    - Enlarged lysosomes and storage granules in all cells
- Abnormal Lysosome and Granules
  - Defective neutrophil function results in defective degranulation and impaired bacteriocidal activity; abnormal chemotaxis
- Impaired NK cell activity
- Melanocytes cannot disperse melanosomes to keratinocytes leading to pigmentation abnormalities
- Platelets have decreased platelet-dense bodies – bleeding diatheses

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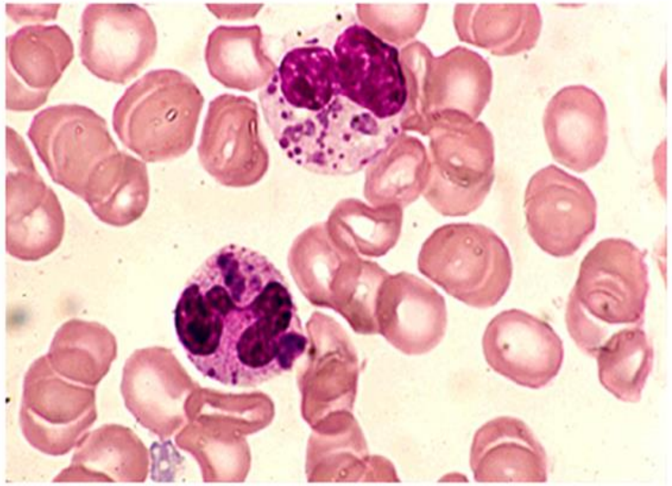
Giant neutrophil granules in Chediak-Higashi syndrome



Peripheral blood smear from a patient with Chediak-Higashi syndrome shows giant granules in the cytoplasm of both a neutrophil and a band form. These granules are formed by the inappropriate fusion of lysosomes and endosomes.

Courtesy of Robert L Baehner, MD. UpToDate®

https://www-uptodate-com.nyit.idm.oclc.org/contents/chediak-higashi-syndrome?search=chediak%20higashi%20syndrome%20children&source=search_result&selectedTitle=1%7E36&usage_type=default&display_rank=1

Chédiak-Higashi Syndrome

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Clinical Features: Frequent and Severe infections

Infectious: Gram + and Gram – bacteria, fungi

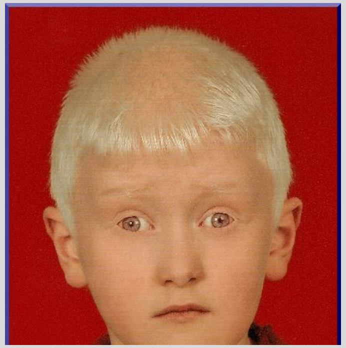
Common Organism

- *Staph. aureus* -most common
- *Streptococcus pyogenes*
- *Pneumococcus*

Site of infection:

- Mucous Membranes
- Skin – superficial or deep abscesses
- Respiratory tract

https://askhematologist.com/chediak-higashi-syndrome/



Non-infectious:

- Partial oculocutaneous albinism [light skin, silvery hair, photophobia, sensitivity]
- Easy bruisability or bleeding diatheses
- Nystagmus
- Peripheral Neuropathy – sensory or motor
- Ataxia
- Can progress to an accelerated phase
  - Pancytopenia, fever, massive lymphohistiocytic infiltration of liver, spleen, nodes
  - Lethal if not treated

https://medlibes.com/entry/chediak-higashi-syndrome

Chédiak-Higashi Syndrome

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Labs: Primary test is genetic testing for CHS1/LYST gene

- Peripheral smear can find giant cytoplasmic granules in leukocytes and platelets
- Neutropenia
- Often present with hypergammaglobulinemia due to frequent infections
- Abnormal platelet aggregation, bleed time

Treatment:

- Prophylactic antibiotics (Trimethoprim-Sulfamethoxazole)
- Aggressive treatment of infections
- IFN -ɣ helps with NK cell function
- G-CSF (Granulocyte colony stimulating factor) corrects neutropenia and decreases infection
- HSCT (hematopoietic stem cell transplant) is the treatment of choice
  - Will prevent accelerated phase
  - Does not prevent neurologic degeneration or oculocutaneous albinism

Chronic Granulomatous Disease (CGD)

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Defect: Defective Phagocytes with inability to form respiratory burst to kill ingested catalase-positive bacteria or fungi

- Transmission: X-linked (65%) or Autosomal Recessive (35%)
  - Involve genes responsible for the NADPH oxidase complex
  - NADPH oxidase complex helps with respiratory/oxidative burst which is an important immune system process for helping phagocytes degrade internalized pathogens.
- In CGD, there is a decreased production of superoxide and H_2O_2 [respiratory burst] so microbes not killed

Clinical Features:

- Characterized by recurrent and severe bacterial and fungal infection with catalase + organism and granulomata formation
  - Think of organs that are barriers from infection [skin, GI tract, lungs, lymph nodes, spleen]
- Usually presents in first 5 years of life

Chronic Granulomatous Disease (CGD)

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Infectious – usually focal but may see bacteremia and fungemia when local infections not treated appropriately

- Types of infections: Pneumonia, Lymphadenitis, Osteomyelitis, Skin
- Catalase-positive Bacterial organisms
  - *Staph aureus* is most common, *Serratia marcesens,* Burkholderia cepacia, *Nocardia, Pseudomonas, Salmonella*
- Catalase-positive Fungal organisms
  - *Aspergillus, Candida albicans*

Non-infectious – Growth failure, Anemia of chronic disease, and **Granulomas**

- Granulomas may be the presenting symptom and occur anywhere
- Pyloric obstruction, bladder outlet obstruction, rectal fistula, granulomatous colitis

Chronic Granulomatous Disease (CGD)

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Labs:

- Neutrophile Function Testing
  - Dihydrorhodamine (DHR)123 Test – Flow cytometry test; Sensitive and Quantitative Test
    - Measures oxidant production showing increased fluorescence when DHR is oxidized via NADPH oxidase pathway
    - Preferred as can distinguish between X-linked and autosomal forms
  - Nitroblue Tetrazolium (NBT) test: - Qualitative test
    - Simple and rapid determination of phagocyte NADPH oxidase activity.
    - NBT is reduced by superoxide and changes from yellow to dark blue in normal cells.
    - X-linked carriers can be identified with this test.
- Genetic Testing
- Other Lab Findings: Hypergammaglobulinemia, Elevated ESR or CRP, Anemia, Hypoalbuminemia

Treatment:

- Aggressively treat active infections: may be more difficult if deep-seated and abscesses may require surgical drainage
- Prophylatic treatment with Trimethoprim-sulfamethoxazole, Itraconzole, and IFN-ɣ
- HSCT (hematopoietic stem cell transplant) is curative
- Gene therapy is being researched

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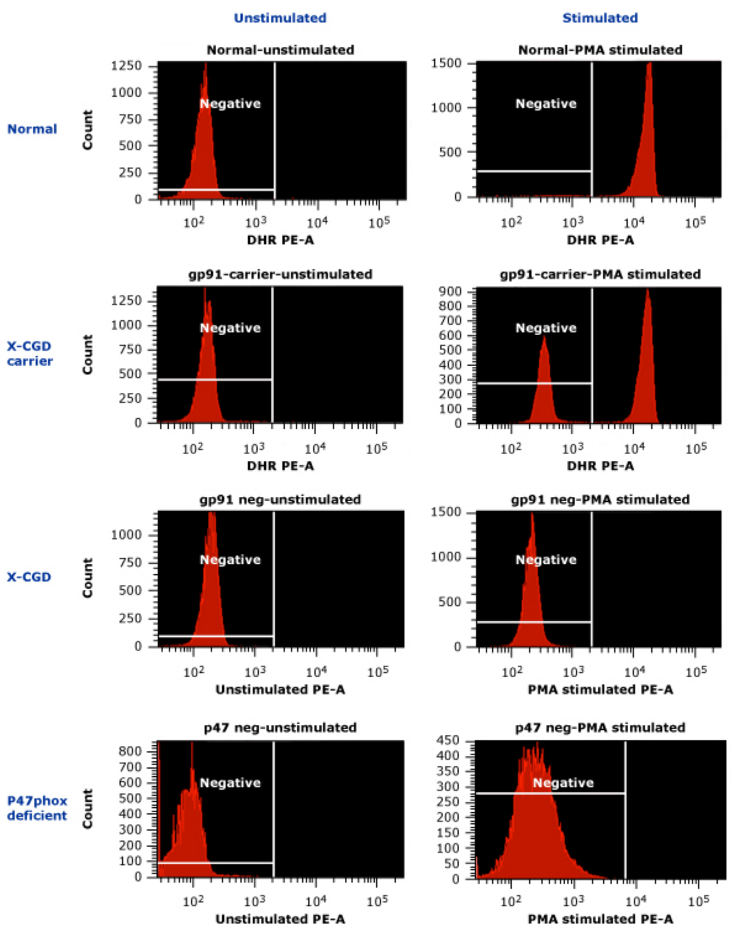
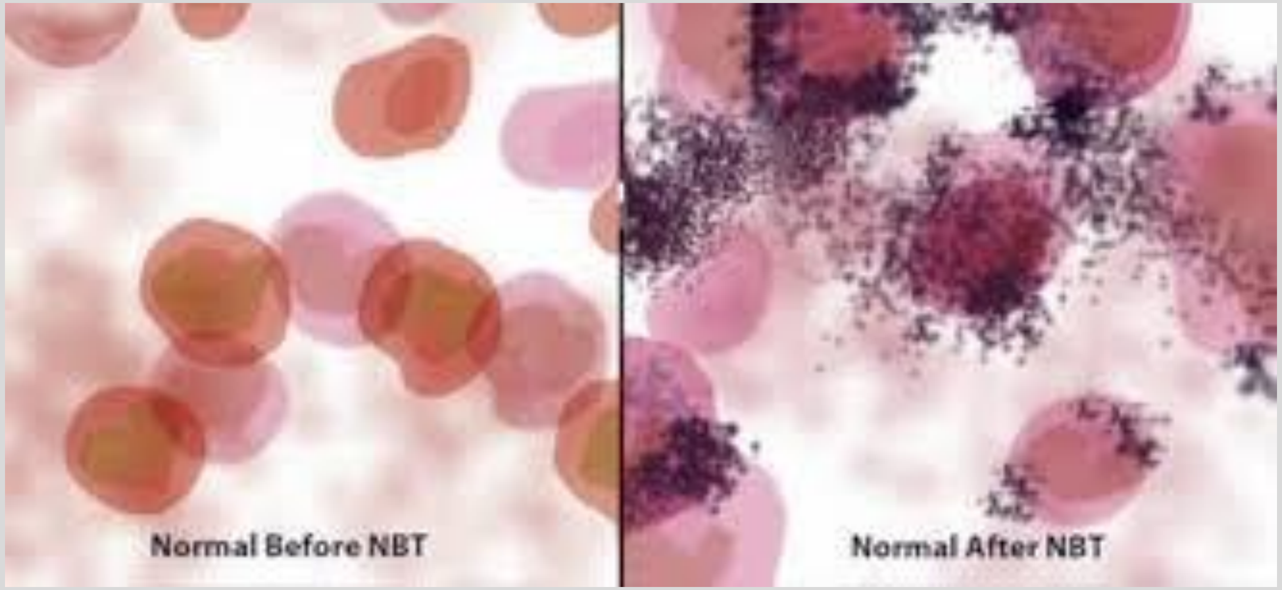
DHR test



DHR assay for CGD diagnosis. Unstimulated (left) and PMA-stimulated (right) neutrophils are shown. Y-axis is number of events; X-axis is fluorescence intensity shown on a log scale.

- In the normal (top panels), there is a rightward shift seen with stimulation.
- In the X-linked CGD carrier state (second row), there are two populations seen: one is normally fluorescent, while the other is essentially the same as the unstimulated population.
- In the patient with X-CGD (third row) there is no shift in fluorescence with stimulation.
- In the patient with the autosomal-recessive p47phox deficiency (fourth row), the rightward shift seen with stimulation is abnormally broad and not very bright.

Up to Date



https://step1.medbullets.com/immunology/105009/chronic-granulomatous-disease

Question 3

A 15 year old male presents the office with his parents with concerns for recurrent bacterial pneumonia. He also has complaints of decreased sensation in his lower extremity. Physical exam reveals, decreased breath sounds and rales in the LLL. He also has very light skin and has greyish silver-sheen hair. What is the most likely associated lab finding?

A. Abnormal nitroblue tetrazolium test

B. Abnormal DHR 123 test

C. Decreased IgG, IgA and IgM

D. Giant granules in granulocytes on peripheral blood smear

Question 3

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Question 4

A 21-year-old female presents with past medical history of autoimmune thrombocytopenia and recurrent sinusitis throughout childhood/adolescence, presents to for follow-up after a recent hospitalization for pneumonia. She has previously been hospitalized for pneumonia. On review of systems, she endorses chronic diarrhea and denies fever, chills and weight loss. X-ray on admission showed non-caseating granuloma in the R and L lungs. Laboratory analysis shows decreased IgG, IgM and IgA. What is the most likely diagnosis?

A. Chronic Mucocutaneous Candidiasis

B. Common variable Immunodeficiency

C. Wiskott Aldrich Syndrome

D. Selective IgA Deficiency

Question 4

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A 21-year-old female presents with past medical history of autoimmune thrombocytopenia and recurrent sinusitis throughout childhood/adolescence, presents to for follow-up after a recent hospitalization for pneumonia. She has previously been hospitalized for pneumonia. On review of systems, she endorses chronic diarrhea and denies fever, chills and weight loss. X-ray on admission showed non-caseating granuloma in the R and L lungs. Laboratory analysis shows decreased IgG, IgM and IgA. What is the most likely diagnosis?

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C. Wiskott Aldrich Syndrome

D. Selective IgA Deficiency

Summary

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- Immunodeficiency disorders involve an impaired and dysregulated immune system, and can be caused by a variety of factors – both primary and secondary
- PIDs are associated with both infections and noninfectious features
- The infections associated with PIDs are recurrent, severe and/or characterized by unusual organisms
- Specific pathogens are associated with specific underling immune defects; knowing these associations can give a clue to the underlying defect
- B cell disorders are associated with bacterial infections of the sinopulmonary tract
- T cell disorders are associated with opportunistic infections with bacteria, fungi and viruses
- Phagocytic disorders are associated with bacterial and fungal infections

Session Objectives

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1. Explain the difference between primary and secondary immunodeficiency (ID) and outline some causes of secondary ID.
2. Explain the signs and symptoms that should raise suspicion for a primary ID (PID).
3. Identify the history and physical exam components important when considering an ID.
4. For the following PIDs, explain the underlying immune defect, the infectious clinical features, the non-infectious clinical features, laboratory features and the main treatment approaches:
    - Common Variable Immunodeficiency
    - Selective IgA Deficiency
    - IL-12 Receptor Deficiency
    - Chronic Mucocutaneous Candidiasis
    - Chédiak-Higashi Disease
    - Chronic Granulomatous Disease

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Lecture Feedback Form:

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Adam Hurst, M.D.
ahurst@nyit.edu

Office hours are by appointment – please email to set a time